Weekly Progress Report

Setterlun University

Date: April 30, 2026

Project: Setterlun University Development

EXECUTIVE SUMMARY

This week, we focused on enhancing user interactivity, optimizing visual transitions between key areas, and establishing the groundwork for a robust Firebase-backed infrastructure. Significant progress was made on the "Claim Brick" functionality and scene transition logic, ensuring a seamless and performant experience for users. We are currently in the process of establishing and testing the connection with Firebase.

1. CORE FUNCTIONALITY ENHANCEMENTS

Claim Brick Logic & Interaction

- Trigger-Based Highlighting: Bricks now dynamically highlight when a user enters their interaction trigger area, providing immediate visual feedback. Highlights are automatically disabled upon exiting the area.
- Form Integration: The interaction flow now correctly handles the state before and after form completion.
- Persistence Feedback: Upon submitting data and returning to the trigger area, the system now displays the specific brick details, confirming successful data capture and persistence.

Lobby Transitions & Scene Management

- Smooth Fade Transitions: Implemented a professional Fade-In/Fade-Out effect when transitioning between the University's main exterior and the Lobby interior.
- Optimized Scene Loading: The transition logic handles scene changes at the backend, ensuring that assets are loaded efficiently without jarring jumps.
- Lighting Optimization: Separated the light baking for exterior and interior (Lobby) scenes. This improves performance by reducing the lightmap overhead and allows for higher visual fidelity.

2. VISUAL & PERFORMANCE OPTIMIZATION

Lighting Fidelity

- Interior & Exterior Baking: Refined the light baking effects to ensure a consistent aesthetic while maintaining high performance.
- Seamless Transitions: Fine-tuned the transition timing so that the shift between different lighting environments feels natural and integrated.

3. QUEST & ONBOARDING DEVELOPMENT

Personal Information Quest

• Phase 1 Implementation: Successfully added "Step A" of the Personal Information Quest.

• Local Storage: Initial implementation is stored locally to facilitate rapid testing and iteration before final backend synchronization.

4. INFRASTRUCTURE & BACKEND INTEGRATION

Firebase Integration Settings

• Backend Toggle: Introduced a BackendConfig option that allows the development team to switch seamlessly between a Custom API and Firebase backend environments.

WebGL-Firebase Bridge (Technical Achievement)

Since Unity does not provide a native Firebase plugin for WebGL, we have implemented a custom JavaScript Bridge. This architecture is critical for WebGL compatibility:

• Unity to Browser: When an action (like Login) is triggered in C#, the bridge passes the request to the browser's native JavaScript environment.

• Browser to Firebase: JavaScript handles the asynchronous communication with Firebase servers directly.

• Browser back to C#: Once Firebase responds, the bridge passes the results back into Unity's C# environment.

Without this bridge, Firebase authentication and database features would be inaccessible in WebGL builds.

5. NEXT WEEK'S ROADMAP

• Finalize Connection: Complete the establishment and testing of the live connection with Firebase.

• Data Synchronization: Once the connection is verified, we will proceed with saving and retrieving data, including User Credentials, Brick Ownership, Avatar Profiles, and Onboarding Quest Data.

End of Report